Basic guidelines on RouterOS configuration and debugging

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! "ril #$%&

' hat is the ain idea of this(

)Little things atter and are very i "ortant*

RouterOS is the sa e every+here

RouterOS anage ent tools

RouterOS anage ent

- CL, -Co and Line ,nterface.

  htt"s/00+iki1 ikrotik1co 0+iki0Manual/Console

- ' ebfig

  htt"s/00+iki1 ikrotik1co 0+iki0Manual/ ' ebfig

- Tik ! ""

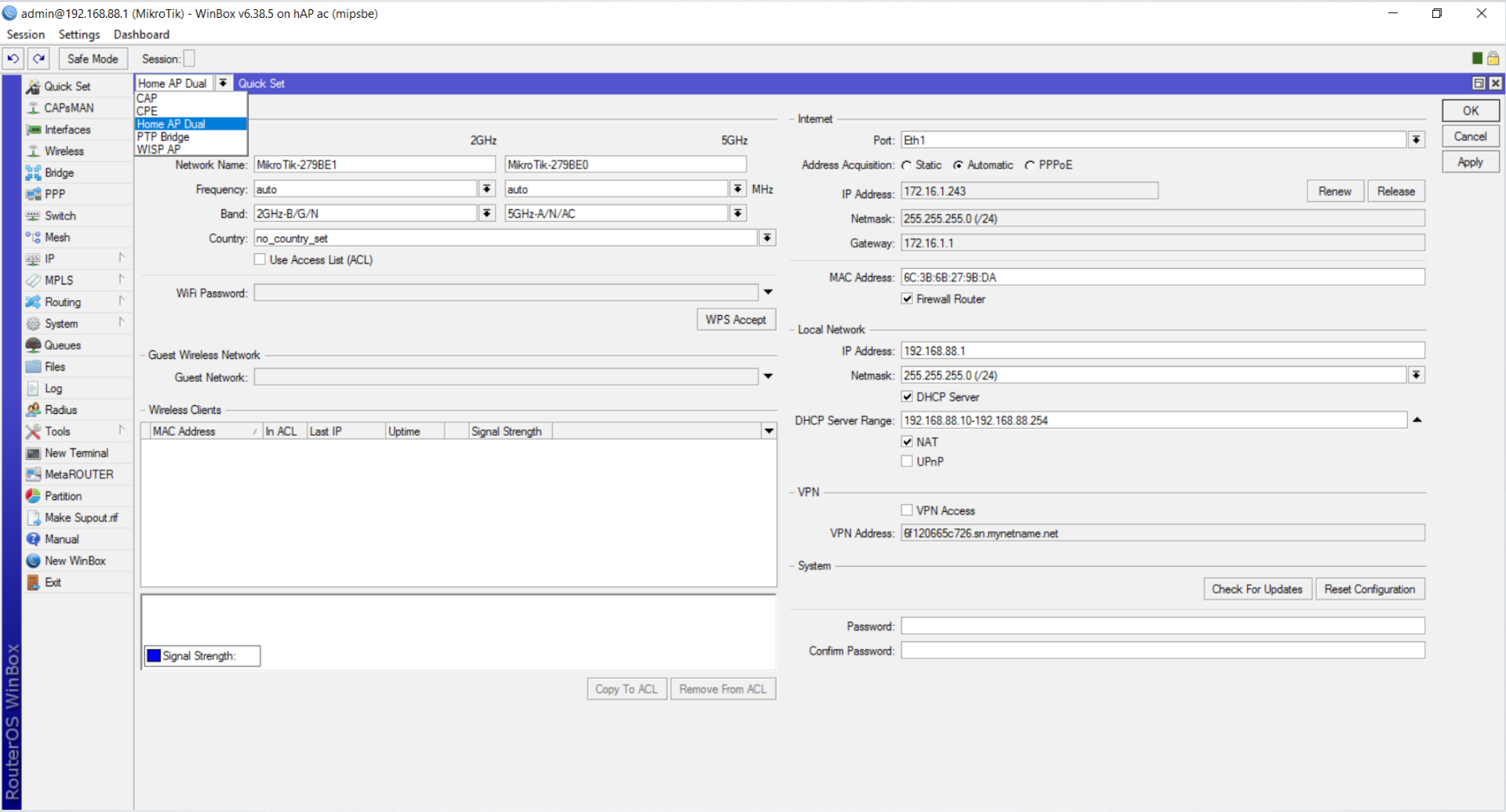
  htt"s/00foru 1 ikrotik1co 0vie+to"ic1"h"(t2345$&

- ' inbo6

  htt"s/00+iki1 ikrotik1co 0+iki0Manual/ ' inbo6

The fastest +ay ho+ to configure device

7 uickSet


admin@192.168.88.1 (MikroTik) - WinBox v6.38.5 on hAP ac (mipsbe)
Session Settings Dashboard
Safe Mode
Session:
Quick Set
Home AP Dual
CAP
CPE
Home AP Dual
PTP Bridge
WISP AP
2GHz
5GHz
Network Name:
MikroTik-279BE1
MikroTik-279BE0
Frequency:
auto
auto
MHz
Band:
2GHz-B/G/N
5GHz-A/N/AC
Country:
no_country_set
Use Access List (ACL)
WiFi Password:
WPS Accept
Guest Wireless Network
Guest Network:
Wireless Clients
MAC Address
In ACL
Last IP
Uptime
Signal Strength
Signal Strength:
Copy To ACL
Remove From ACL
Internet
Port:
Eth1
Address Acquisition:
Static
Automatic
PPPoE
IP Address:
172.16.1.243
Renew
Release
Netmask:
255.255.255.0 (/24)
Gateway:
172.16.1.1
MAC Address:
6C:3B:6B:27:9B:DA
Firewall Router
Local Network
IP Address:
192.168.88.1
Netmask:
255.255.255.0 (/24)
DHCP Server
DHCP Server Range:
192.168.88.10-192.168.88.254
NAT
UPnP
VPN
VPN Access
VPN Address:
6f120665c726.sn.mynetname.net
System
Check For Updates
Reset Configuration
Password:
Confirm Password:
OK
Cancel
Apply
Quick Set
CAPsMAN
Interfaces
Wireless
Bridge
PPP
Switch
Mesh
IP
MPLS
Routing
System
Queues
Files
Log
Radius
Tools
New Terminal
MetaROUTER
Partition
Make Supout.rif
Manual
New WinBox
Exit
RouterOS WinBox

7 uickSet

- 8asy to use
- Contains the ost co only used features and should be enough for basic usage

9olden rule about 7 uickSet/

),f you use 7 uickSet, then use 7 uickSet, if you leave it, then forget about it111*

Si "le security

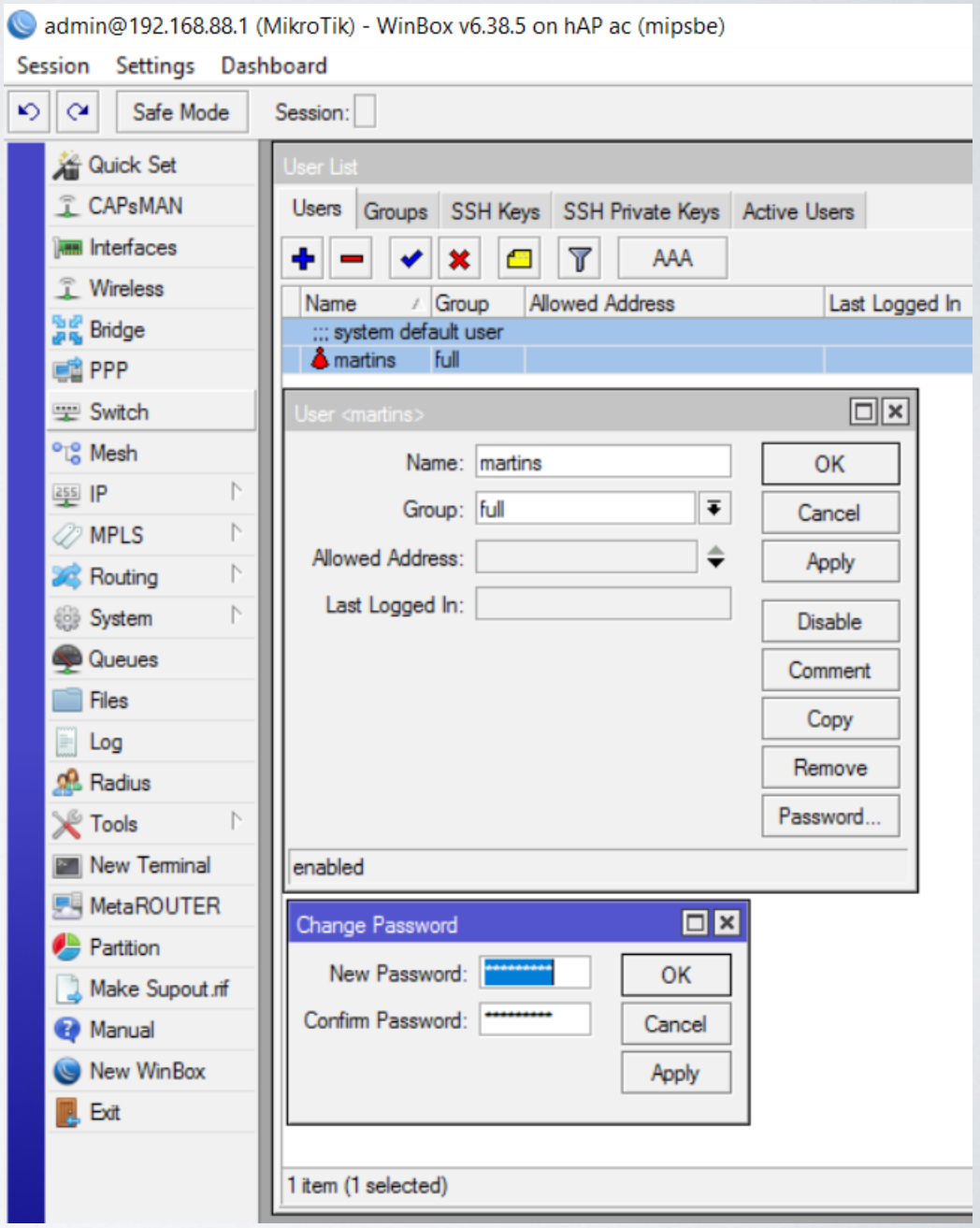
Si "le security

- S"ecify user "ass+ord

  0user set ad in "ass+ord2:::

- ; se different userna e

  0user set ad in na e2 artins

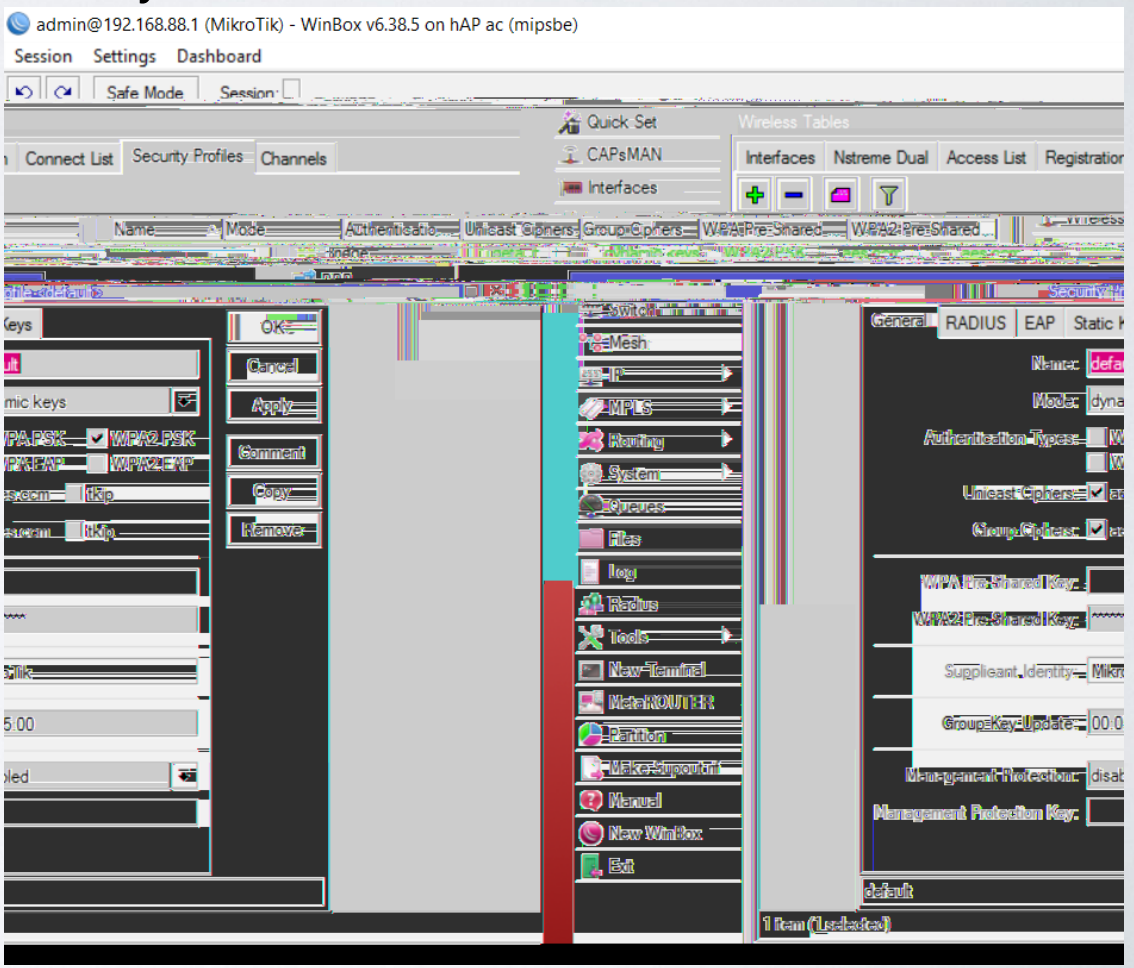


Si "le security

- S"ecify "ass+ord for +ireless access

  0interface +ireless security<"rofiles set default authentication<ty"es2+"a#<"sk

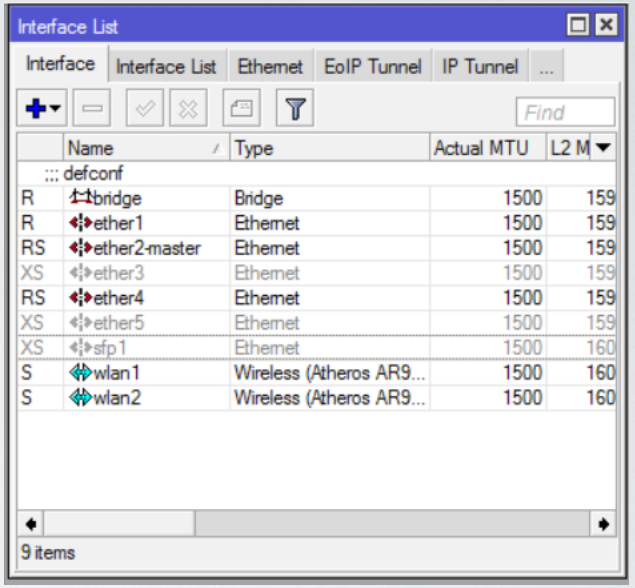
  ode2dyna ic<keys +"a#<"re<shared<key2:::::::



Si "le security

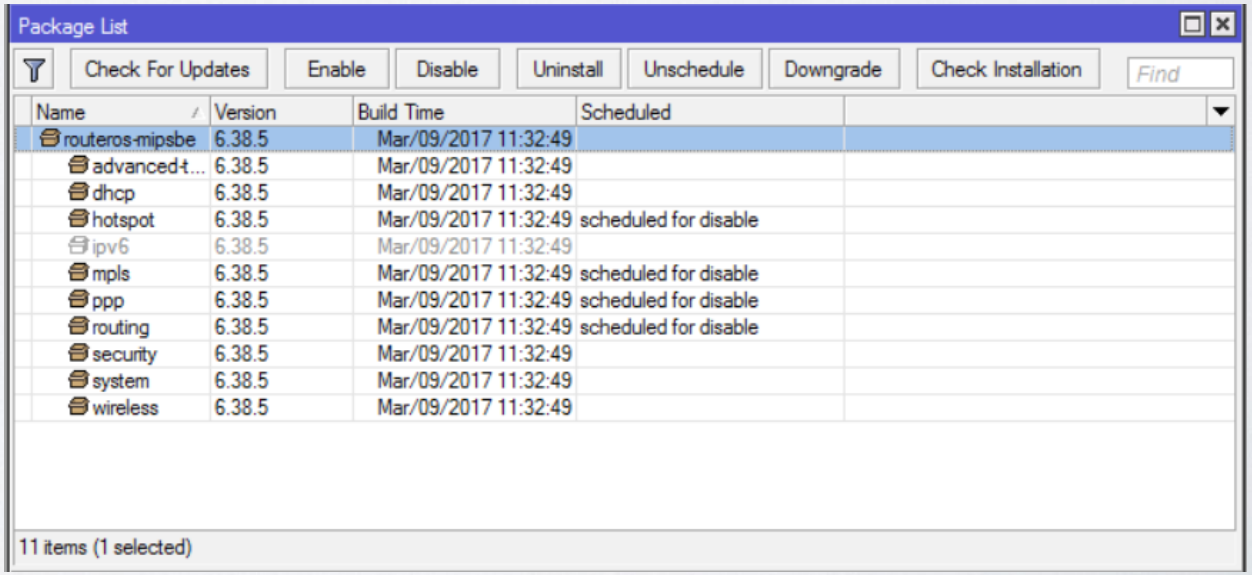
- =isable unused interfaces

  0interface ethernet disable ether>,ether?,sf"%



- =isable unused "ackages - ainly ,@vA.

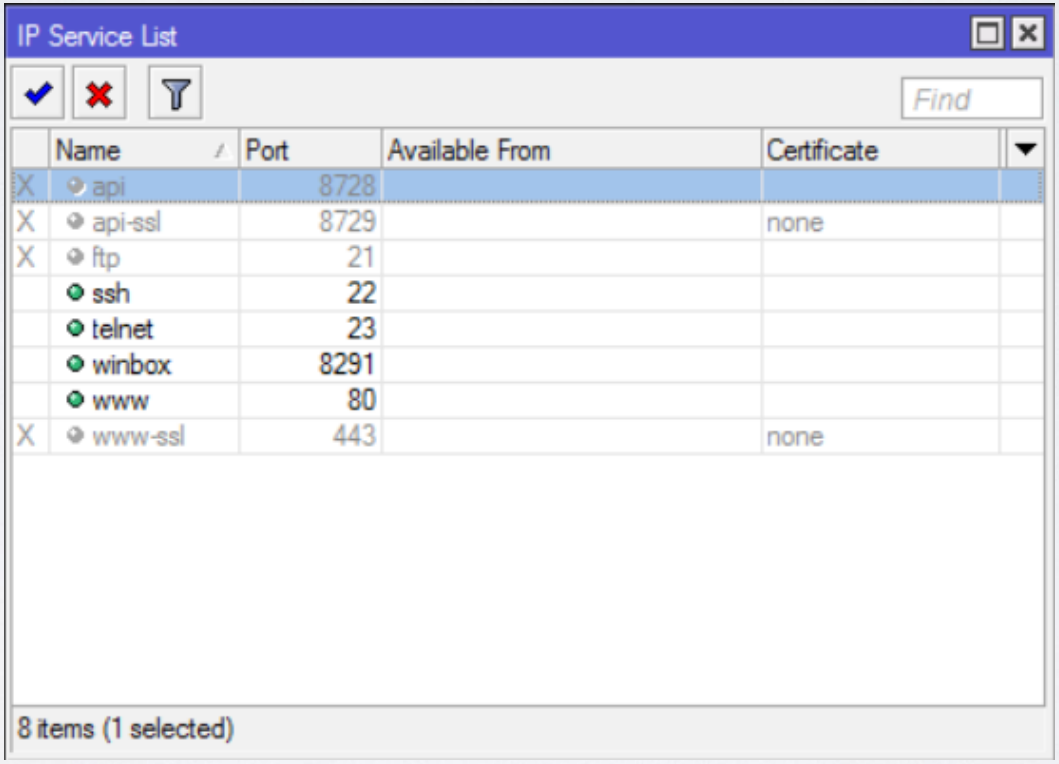
  0syste "ackage disable hots"ot,i"vA, "ls,"""",routing



Si "le security

- =isable ,@0Services

  0i" service disable a"i,a"i<ssl,ft",+++<ssl



Si "le security

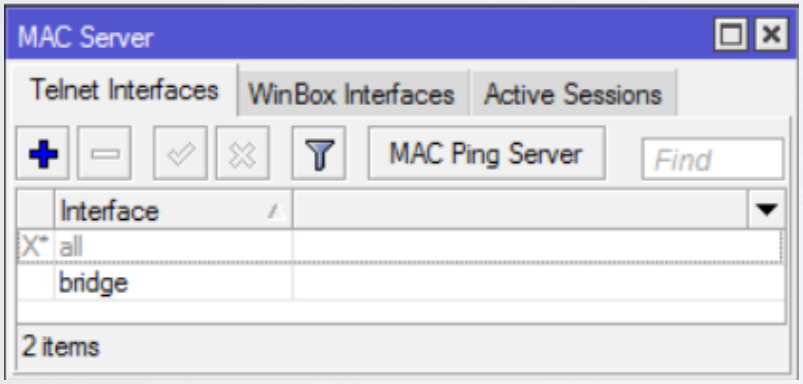
- ! dBust M ! C access

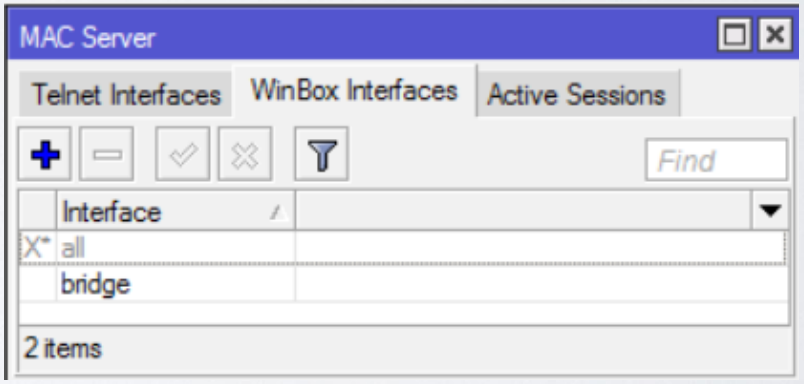
  0tool ac<server set C find default2yes D disabled2yes

  0tool ac<server add interface2bridge

  0tool ac<server ac<+inbo6 set C find default2yes D disabled2yes

  0tool ac<server ac<+inbo6 add interface2bridge

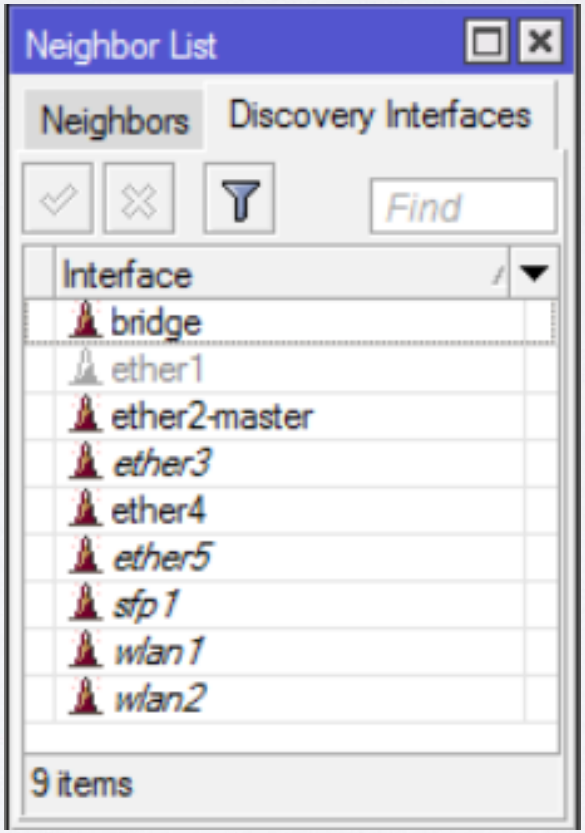




Si "le security

- Hide device in Eeighbor =iscovery

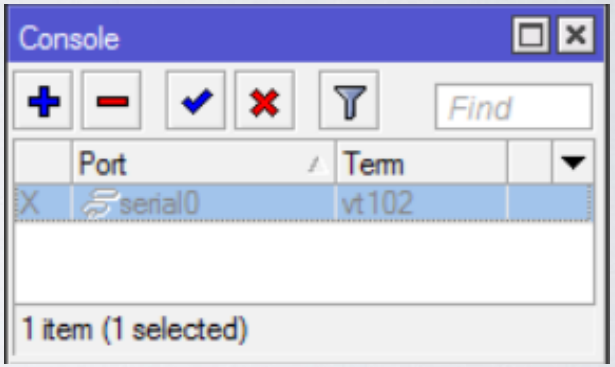
  0i" neighbor discovery set ether% discover2no



Si "le security

- =isable serial "ort if not used -and if included.

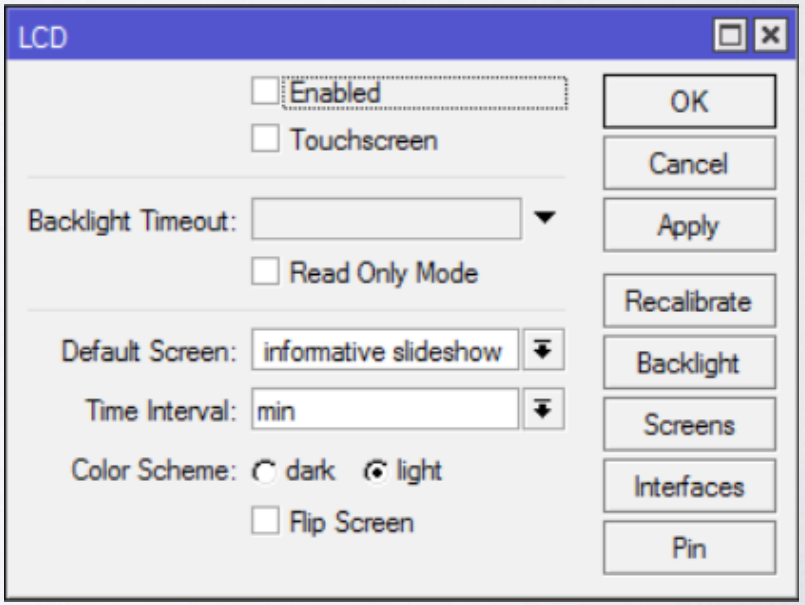
  0syste console disable Cfind +here "ort2serial$D



- =isable LC=

  0lcd set enabled2no

  0lcd set touch<screen2disabled



Si "le security

- @rotect reset button

  0syste routerboard settings set "rotected<routerboot2enabled refor at<hold<button2>$s

  htt"s/00+iki1 ikrotik1co 0+iki0Manual/RouterBO ! R=FsettingsG@rotectedFbootloader

Hire+all

- Hair"in E ! T -access local resource through "ublic ,@.

  htt"s/00+iki1 ikrotik1co 0+iki0Hair"inFE ! T

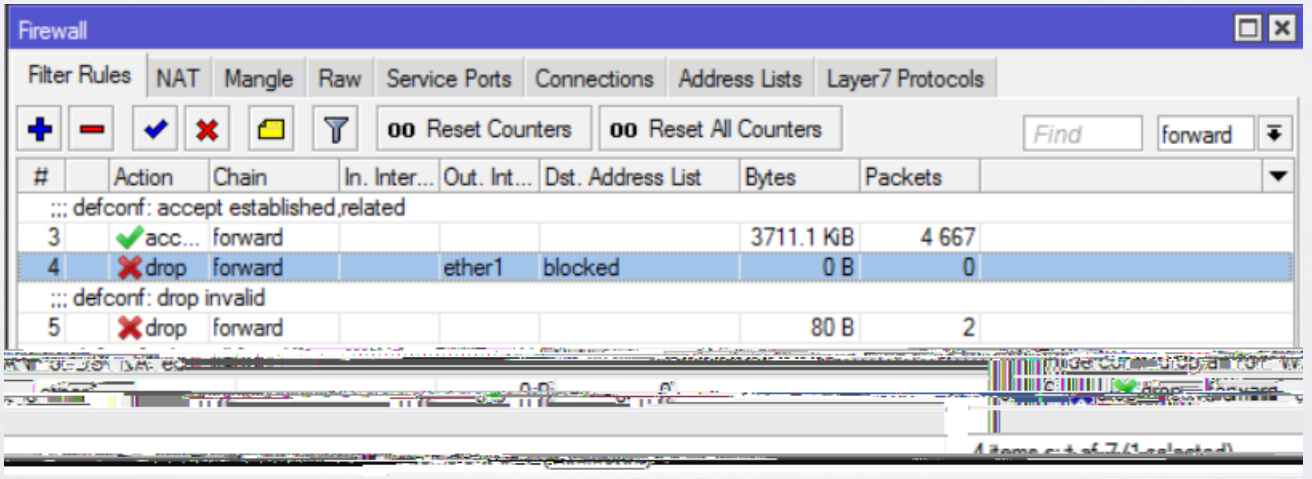
Hire+all

- Block s"ecific traffic

  0i" fire+all address<list add list2blocked address2+++1facebook1co

  0i" fire+all filter add chain2for+ard action2dro" dst<address<list2blocked out<

  interface2ether%



Hire+all

- @rotect device against attacks, if you allo+ "articular access

  0i" fire+all filter

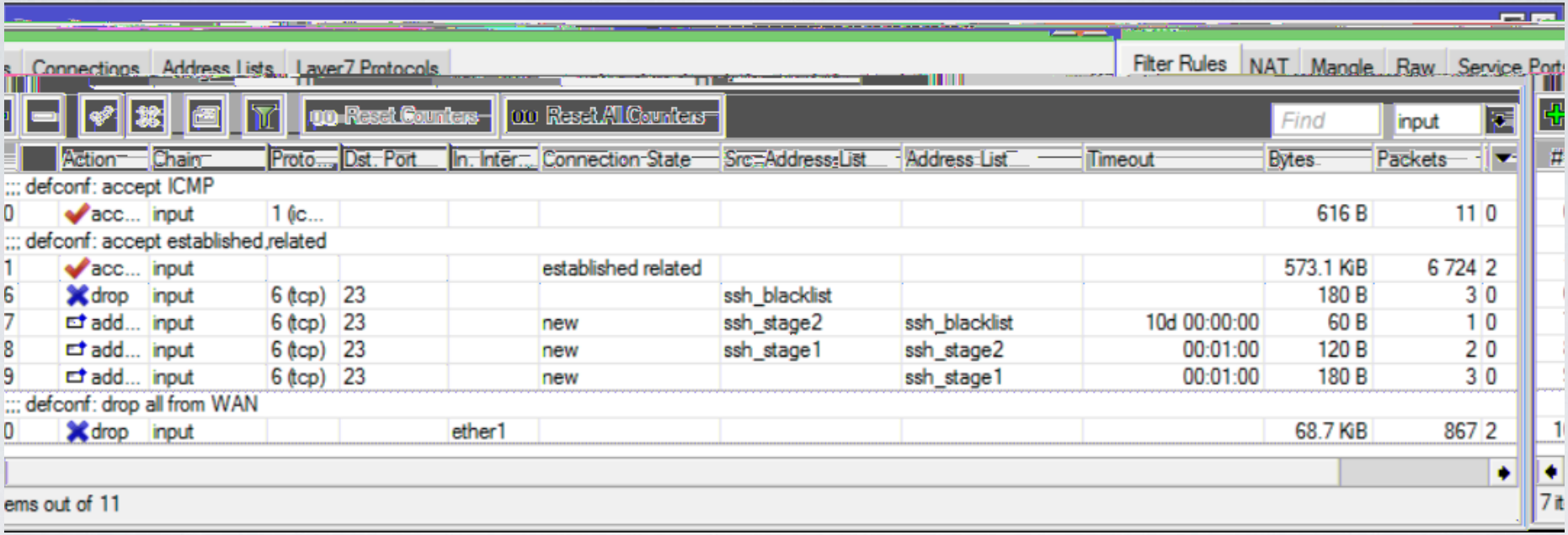
  add chain2in"ut "rotocol2tc" dst<"ort2#> src<address<list2sshFblacklist action2dro"

  add chain2in"ut "rotocol2tc" dst<"ort2#> connection<state2ne+ src<address<list2sshFstage# action2add<src<to<address<list address<list2sshFblacklist address<list<ti eout2%$d

  add chain2in"ut "rotocol2tc" dst<"ort2#> connection<state2ne+ src<address<list2sshFstage% action2add<src<to<address<list address<list2sshFstage# address<list<ti eout2%

  add chain2in"ut "rotocol2tc" dst<"ort2#> connection<state2ne+ action2add<src<to<address<list address<list2sshFstage% address<list<ti eout2%

Hire+all



htt"s/00+iki1 ikrotik1co 0+iki0BruteforceFloginF"revention

Handle band+idth

HastTrack

- Re e ber this rule(

  0i" fire+all filter

  add chain2for+ard action2acce"t connection<state2established,related

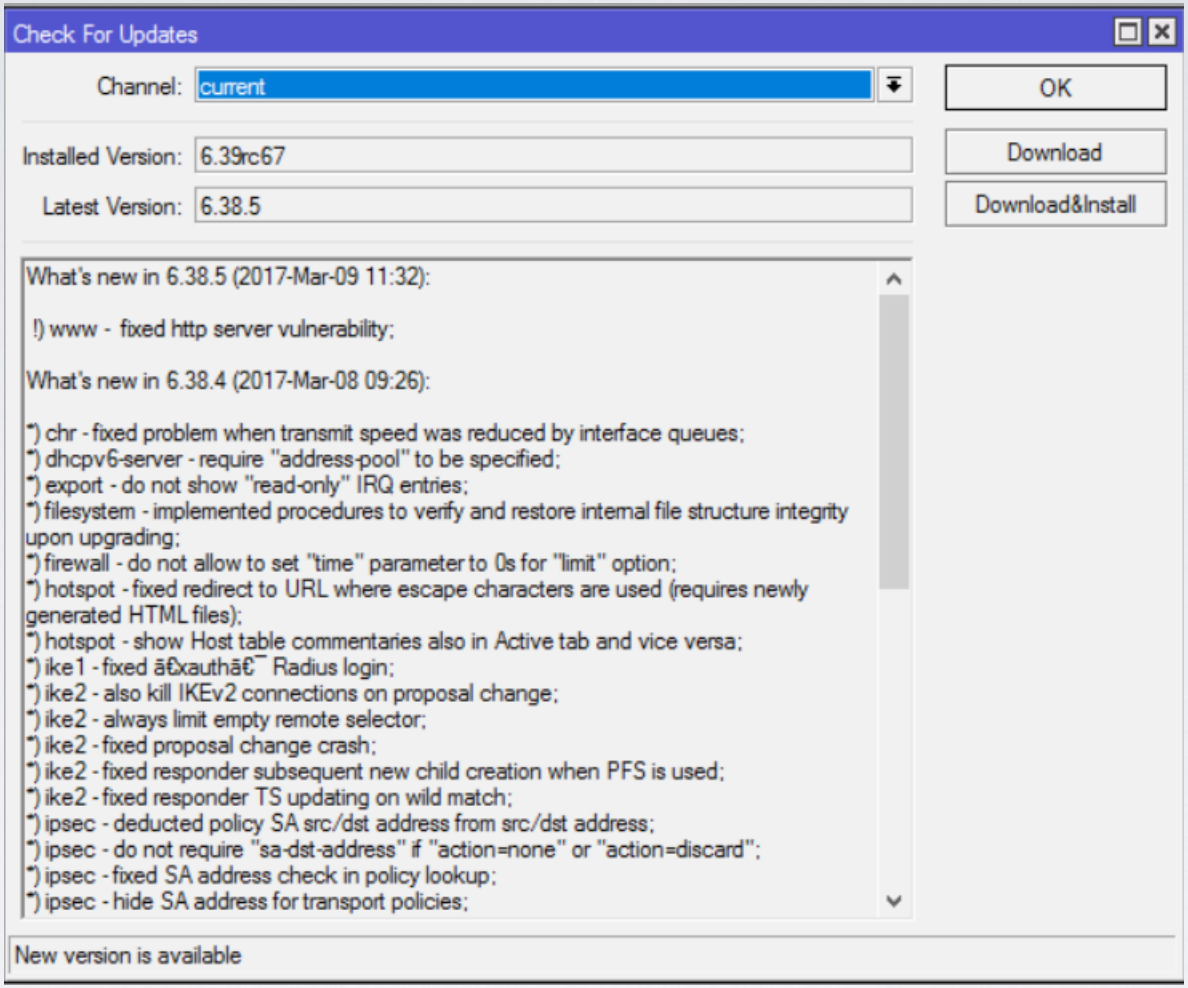
- !dd HastTrack rule before "revious one

  0i" fire+all filter

- add chain2for+ard action2fasttrack<connection connection<

  state2established,related

; "grade device



htt"s/00+iki1 ikrotik1co 0+iki0Manual/ ; "gradingFRouterOS

' hat to do +hen soft+are sto"s +orking(

Resolve "roble s

- Backu" RouterBOOT

  %. @o+er device off, "ress and hold the reset button

  #. @o+er device on and after %<# seconds release the button

- Eetinstall

  %. Test Eetinstall

  htt"s/00+iki1 ikrotik1co 0+iki0Manual/Eetinstall

  #. Try to re<install any other router

- Reset device

  htt"s/00+iki1 ikrotik1co 0+iki0Manual/Reset

Resolve "roble s

- Serial "ort

  %. Sho+s all available infor ation -also booting.

  #. ' ill +ork if "roble is related to Layer#0Layer>

  connectivity and0or interfaces the selves

- 86change device
- Choose ore "o+erful device -or ulti"le devices.

, can not figure it out by yself

Configuration issues

- Consultants0=istributors/
  - htt"s/00 ikrotik1co 0consultants
  - htt"s/00 ikrotik1co 0buy
- !sk for hel" in foru /
  - htt"s/00foru 1 ikrotik1co 0
- Look for an ans+er in anual
  - htt"s/00+iki1 ikrotik1co 0+iki0MainF@age